

Tasks on EC2

1. Launch an EC2 instance using the Amazon Linux 2 image and add a script in user data to install Apache HTTPD.

Steps to Launch an instance with Amazon Linux 2 AMI and install HTTPD at the time of installation:

Step 1: Open the AWS website and navigate to the EC2 Service

- Log in to AWS Management Console
- Search for "EC2" and select the service
- Click "Launch Instance"

Step 2: Configure Basic Settings

- Name: Enter "WebServer-AmazonLinux2-Apache"
- AMI Selection: Choose "Amazon Linux 2 AMI (HVM) - Kernel 5.10"
- Instance Type: Select "t2.micro" (eligible for free tier)

Step 2: Configure Basic Settings

- Name: Enter "WebServer-AmazonLinux2-Apache"
- AMI Selection: Choose "Amazon Linux 2 AMI (HVM) - Kernel 5.10"
- Instance Type: Select "t2.micro" (eligible for free tier)

Step 3: Configure Key Pair

- Select an existing key pair or create a new one
- This enables SSH access to your instance

Step 4: Network Settings

- Create or select a security group
- Add rules for HTTP (port 80) and SSH (port 22)
- Source: "0.0.0.0/0" for HTTP, your IP for SSH (more secure)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

AMI from catalog

Quick Start

Name

amzn2-ami-hvm-2.0.20250902.3-x86_64-gp2

Verified provider

Description

Amazon Linux 2 AMI 2.0.20250902.3 x86_64 HVM gp2

Image ID

ami-0023921b4fcd5382b

Username

ec2-user

Catalog

AWS Marketplace AMIs

Published

2025-09-03T05:26:54.000Z

Architecture

x86_64

Virtualization

hvm

Root device type

ebs

ENA Enabled

Yes

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

▼ Summary

Number of instances [Info](#)

1

Software Image (AMI)

Amazon Linux 2 AMI (HVM), SSD ...[read more](#)

ami-0023921b4fcd5382b

Virtual server type (instance type)

t3a.medium

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 30 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the

Step 5: Add User Data Script. In the "Advanced details" section, paste this user data script:

```
#!/bin/bash
# yum update -y – Updating system
# yum install httpd -y – Installing httpd
# systemctl start httpd – Starting httpd
# systemctl enable httpd – Creating a symlink
# echo "Context" > /var/www/html/index.html – Adding content into the file.
# chown -R apache:apache /var/www/html/index.html – Giving ownership to deploy the file.
```

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum update -y
yum upgrade -y
yum install httpd -y
systemctl start nginx
systemctl enable nginx
echo "15 minutes is all it takes" > /var/www/html/index.html
echo "Launched instance on 18th of Sept, 6:45 p.m." >>
/var/www/html/index.html
chown -R apache:apache /var/www/html/
```

☐ User data has already been base64 encoded

new security group

Storage (volumes)
1 volume(s) - 30 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

[Cancel](#) [Launch instance](#) [Preview code](#)

Step 6: Launch Instance

- Review all settings and click "**Launch Instance**"
- Check if the instance is initialized properly. When the status check is passed, the instance is now successfully active and ready to use.
- You may now connect to the instance using any tool. I will be using Git Bash.

Success
Successfully initiated launch of instance (i-0c1676814d23c5c03)

▼ Launch log

Initializing requests	✔ Succeeded
Creating security groups	✔ Succeeded
Creating security group rules	✔ Succeeded
Subscribing to Marketplace AMI	✔ Succeeded
Launch initiation	✔ Succeeded

Instances (1/1) [Info](#) Last updated less than a minute ago [Refresh](#) [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Find Instance by attribute or tag (case-sensitive) Running

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☒	AmazonLinux2	i-0c1676814d23c5c03	Running	t3a.medium	3/3 checks passed	View alarms	us-east-1f

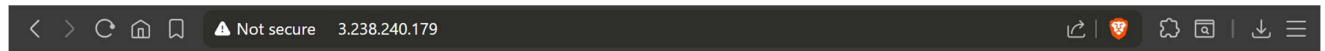
Step 7: Check the result.

- Check if HTTPD is installed successfully by using the following commands:
systemctl status httpd
- Copy the public IP of the instance and open it with the port number 80 in a browser.

- If the results of the deployment are successful, then the web page will show the content we gave it at the time of launching the instance.
- If the script that we gave at the launch of the instance fails, the process will continue to launch the EC2 instance without installing HTTPD. You can manually install HTTPD once you have connected to the instance.

```
[root@ip-172-31-69-77 /]# systemctl status httpd
● httpd.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; vendor preset: disabled)
  Active: inactive (dead)
  Docs: man:httpd.service(8)
[root@ip-172-31-69-77 /]# systemctl start httpd
[root@ip-172-31-69-77 /]# systemctl status httpd
● httpd.service - The Apache HTTP Server
  Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; vendor preset: disabled)
  Active: active (running) since Thu 2025-09-18 13:27:19 UTC; 5s ago
  Docs: man:httpd.service(8)
  Main PID: 5635 (httpd)
  Status: "Processing requests..."
  CGroup: /system.slice/httpd.service
          └─5635 /usr/sbin/httpd -DFOREGROUND
             └─5636 /usr/sbin/httpd -DFOREGROUND
                └─5637 /usr/sbin/httpd -DFOREGROUND
                   └─5638 /usr/sbin/httpd -DFOREGROUND
                      └─5639 /usr/sbin/httpd -DFOREGROUND
                         └─5640 /usr/sbin/httpd -DFOREGROUND

Sep 18 13:27:19 ip-172-31-69-77.ec2.internal systemd[1]: Starting The Apache HTTP Server...
Sep 18 13:27:19 ip-172-31-69-77.ec2.internal systemd[1]: Started The Apache HTTP Server.
[root@ip-172-31-69-77 /]#
```



15 minutes is all it takes Launched instance on 18th of Sept, 6:45 p.m.

The result of the deployed file is attached above.

2. Launch an EC2 using an Ubuntu image and add a script in user data to install Nginx.

Steps to launch EC2 using the Ubuntu image and install and run Nginx while launching the instance:

EC2

Instances

Launch an instance

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-0360c520857e3138f (64-bit (x86)) / ami-026fccd88446aa0bf (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

AMI ID

Publish Date

Username

Verified provider

64-bit (x...

ami-0360c520857e3138f

2025-08-21

ubuntu

Instance type

t2.micro

Free tier eligible

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.0162 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour

On-Demand SUSE base pricing: 0.0116 USD per Hour

On-Demand RHEL base pricing: 0.026 USD per Hour

On-Demand Linux base pricing: 0.0116 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Step 1: Launch New Instance

- Click "Launch Instance" again
- Name: "Ubuntu"

Step 2: Select Ubuntu AMI

- Choose "Ubuntu Server 22.04 LTS (HVM)"
- Instance Type: "t2.micro"

Step 3: Use the same Security Group

- Select the security group created in Task 1 or create similar rules

EC2

Instances

Launch an instance

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

ec2

Create new key pair

Network settings

VPC - required

vpc-07d09424f4ccc9251

(default)

Create new VPC

Subnet

No preference

Create new subnet

Availability Zone

No preference

Enable additional zones

Auto-assign public IP

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups)

Create security group

Select existing security group

Common security groups

Select security groups

MySecurityGroup sg-0a98415f0a592a1bf

Compare security group rules

Step 4: Add Nginx User Data Script

```
#!/bin/bash
```

```
apt-get update -y
```

```
apt-get install -y nginx
```

```
systemctl start nginx
```

```
systemctl enable nginx
```

```
cat > /var/www/html/index.html << 'HTML'
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Ubuntu with Nginx</title>
```

```
</head>
```

```
<body>
```

```
  <h1>Hello from Ubuntu with Nginx!</h1>
```

```
  <p>Nginx is running successfully!</p>
```

```
</body>
```

```
</html>
```

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
cat > /var/www/html/index.html << 'EOF'
<!DOCTYPE html>
<html>
<head>
  <title>Ubuntu with Nginx</title>
</head>
<body>
  <h1>Hello from Ubuntu with Nginx!</h1>
  <p>Server launched on: DATE_PLACEHOLDER</p>
  <p>Nginx is running successfully!</p>
</body>
</html>
EOF
```

[EC2](#) > [Instances](#) > Launch an instance

Success
Successfully initiated launch of instance (i-023825c8795dcd9bd)

▼ Launch log

Initializing requests	✓ Succeeded
Launch initiation	✓ Succeeded

[EC2](#) > [Instances](#)

Successfully initiated termination (deletion) of i-0c1676814d23c5c03,i-0b6fe8b74522d3e53

Instances (1) [Info](#)

Last updated
less than a minute ago

[Connect](#)

[Instance state](#)

[Actions](#)

[Launch instances](#)

[Running](#)

< 1 > [Settings](#)

☐	Name ✎	Instance ID	Instance state ▼	Instance type ▼	Status check ▼	Alarm status ▼	Availability Zone ▼	Public IPv4 DNS
☐	Ubuntu	i-023825c8795dcd9bd	✓ Running 🔍 🔍	t2.micro	✓ 2/2 checks passed View alarms +	us-east-1b	ec2-54-197-31-138.co..	

Step 6: Launch Instance

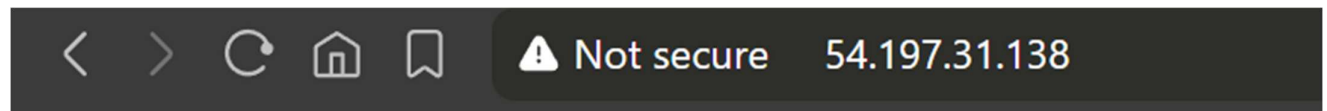
- Review all settings and click "**Launch Instance**"
- Check if the instance is initialized properly. When the status check is passed, the instance is now successfully active and ready to use.
- You may now connect to the instance using any tool. I will be using Git Bash.

```
root@ip-172-31-27-16:/home/ubuntu# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-09-18 20:31:56 UTC; 9min ago
     Docs: man:nginx(8)
   Main PID: 1566 (nginx)
    Tasks: 2 (limit: 1121)
  Memory: 1.8M (peak: 3.7M)
     CPU: 18ms
   CGroup: /system.slice/nginx.service
           └─1566 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─1568 "nginx: worker process"

Sep 18 20:31:56 ip-172-31-27-16 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server:
Sep 18 20:31:56 ip-172-31-27-16 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server:
lines 1-14/14 (END)
```

Step 7: Check the result.

- Check if nginx is installed successfully by using the following commands:
systemctl status nginx
- Copy the public IP of the instance and open it with the port number 80 in a browser.
- If the results of the deployment are successful, then the web page will show the content we gave it at the time of launching the instance.
- If the script that we gave at the launch of the instance fails, the process will continue to launch the EC2 instance without installing nginx. You can manually install nginx once you have connected to the instance.



Hello from Ubuntu with Nginx!

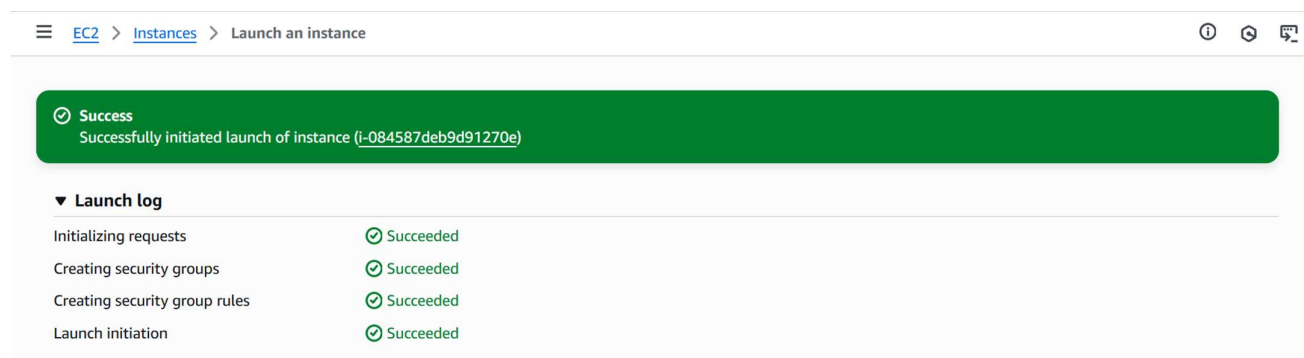
Nginx is running successfully!

Note: There are two deployment locations for Nginx, one is `‘/var/www/html/index.html’`, and the second one is `‘/usr/share/nginx/html/index.html’`. The deployment location may change depending on the flavour of Linux we use. I have used Ubuntu and Amazon Linux to date, and in my observation, it is working in both locations.

3. Launch a Windows server and install Tomcat in Windows.

Steps to launch a Windows Server using the AWS EC2 service, install Tomcat, and deploy a custom sample app.

1. Launch a Windows Server AMI and create/assign a security group that allows RDP (3389), a TCP port (8080) for Tomcat, and an HTTP/HTTPS port 80/443.
2. To connect the machine, click on the instance ID and click on the Connect button.
3. A new page will open upon clicking on **“Connect”**. Click on the **“RDP Client”** section and scroll down, search for **“Password -> Get password”**.



The screenshot displays the AWS Management Console interface for launching an EC2 instance. The breadcrumb navigation at the top reads "EC2 > Instances > Launch an instance". A prominent green success banner states "Success" and "Successfully initiated launch of instance (i-084587deb9d91270e)". Below this, the "Launch log" section is expanded, showing a list of steps that all completed successfully:

Step	Status
Initializing requests	✓ Succeeded
Creating security groups	✓ Succeeded
Creating security group rules	✓ Succeeded
Launch initiation	✓ Succeeded

Instance summary for i-084587deb9d91270e (WindowsServer) Info

Updated less than a minute ago

Instance ID
i-084587deb9d91270e

IPv6 address
-

Hostname type
IP name: ip-172-31-19-71.ec2.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
98.88.84.59 [Public IP]

IAM Role
-

IMDSv2
Required

Operator
-

Public IPv4 address
98.88.84.59 [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-19-71.ec2.internal

Instance type
t3.medium

VPC ID
vpc-07d09424f4ccc9251

Subnet ID
subnet-09a7e3d7f2420cb01

Instance ARN
arn:aws:ec2:us-east-1:734846753465:instance/i-084587deb9d91270e

Private IPv4 addresses
172.31.19.71

Public DNS
ec2-98-88-84-59.compute-1.amazonaws.com [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
[Opt-in to AWS Compute Optimizer for recommendations.](#) | [Learn more](#)

Auto Scaling Group name
-

Managed
false

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance details Info

AMI ID
ami-0e16d075ec2375cf5

Monitoring
disabled

Platform details
Windows

4. Click on **“Get Password”**. Click on **“Upload private key file”** and upload the PEM key, which is used at the time of creating the instance.

5. Click on the **“Decrypt Password”** button to get the password. Use this password to connect to the Windows Server.

6. On the same page, there will be a **“Download remote desktop file”** button. Click and download the file. It will be used to connect to the remote Windows Server. Use the password to log in.

Session Manager | **RDP client** | EC2 serial console

Record RDP connections
You can now record RDP connections using AWS Systems Manager just-in-time node access. [Learn more](#) [Try for free](#) [×](#)

Instance ID
i-084587deb9d91270e (WindowsServer)

Connection Type

☒ **Connect using RDP client**
Download a file to use with your RDP client and retrieve your password.

☐ **Connect using Fleet Manager**
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS
ec2-54-92-211-240.compute-1.amazonaws.com

Username Info
Administrator

Password [Get password](#)

aws Search [Alt+S] United States (N. Virginia) Account ID: 7345-4075-3465 onerfarooq257

EC2 > Instances > i-084587deb9d91270e > Get Windows password

Get Windows password info

Use your private key to retrieve and decrypt the initial Windows administrator password for this instance.

Instance ID
i-084587deb9d91270e (WindowsServer)

Key pair associated with this instance
ec2

Private key
Either upload your private key file or copy and paste its contents into the field below.

[Upload private key file](#)

ec2.pem
1,678KB

Private key contents - optional

```
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEAwe25u2POBcy3l6o5dMqh3R+fkfJd8BsJ8lotfhC2F294KkVl
lbEovY3RQRhV6+HJ6LqRlZB1g/XhonAld4VJz5nou1CnPH6isB9MHfPkRvYzXqX
TE6k+R53mrYUPPrJdt58mwX3YvafkVgkLpKVAalHeKExpI92RgmliOriapVH9
UuDbczJufe2qDahivNUxGn5N8v3ihP6Q3vg7KqNZ0gkGYu8PYfokPn2FVNH33g+
jL013X93Nysdta8KcMs4vQ8xb0OpV+tCsDWGIClhJxp64/LKCRVlnQLF170oNh/
XKwu7Urt2/ertpozWt02gAMcR+ZmORS5h6T+tkwIDAQABAolBAQC/cgObDspCm3jB
NucFyOINy7yp3fPJq8/CMQb8Hw9ET3z3bg6KilPVMe3UL5WJjhs/r38tgbEHlskr
-----
```

[Cancel](#) [Decrypt password](#)

7. On the Windows instance: install Java (JDK/JRE) and then install Tomcat via the Windows service installer.

8. Tomcat requires a Java runtime. Download a Java JDK suitable for the Tomcat version (e.g., Temurin / Eclipse Temurin / OpenJDK). Install it via the MSI or installer. I used Microsoft OpenJDK 21.

WindowsServer - ec2-52-91-58-131.compute-1.amazonaws.com - Remote Desktop Co...

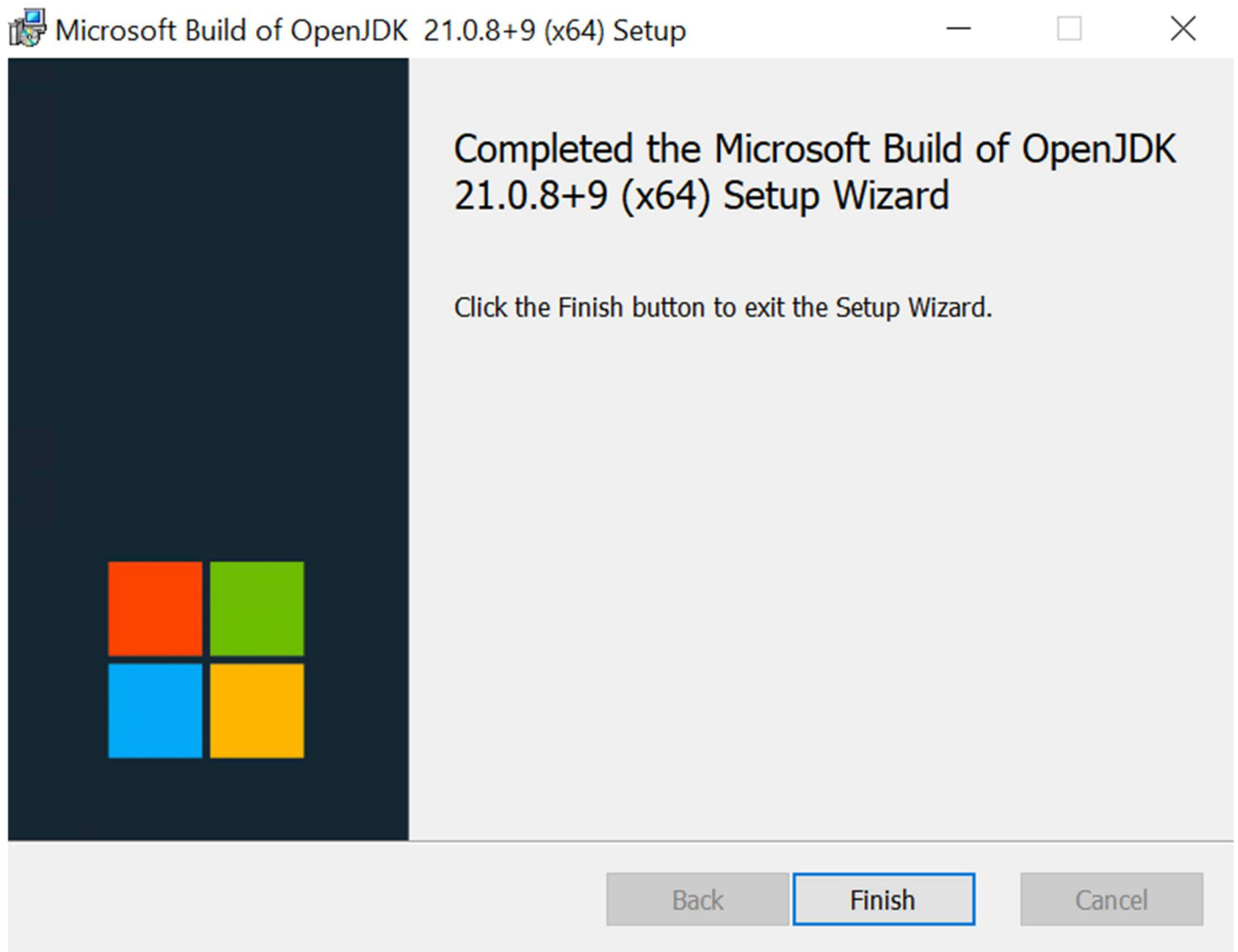
download the Microsoft Build of [Java](#)

java/openjdk/download

OpenJDK 21.0.8 LTS | [See previous releases](#)

[Expand table](#)

Platform	Architecture	Type	Download link	Other files
x64				
Linux	x64	tar.gz	microsoft-jdk-21.0.8-linux-x64.tar.gz i2	sha256 i2 / sig i2
macOS	x64	pkg	microsoft-jdk-21.0.8-macos-x64.pkg i2	sha256 i2
macOS	x64	tar.gz	microsoft-jdk-21.0.8-macos-x64.tar.gz i2	sha256 i2 / sig i2
Windows	x64	msi	microsoft-jdk-21.0.8-windows-x64.msi i2	sha256 i2
Windows	x64	zip	microsoft-jdk-21.0.8-windows-x64.zip i2	sha256 i2 / sig i2
AArch64				



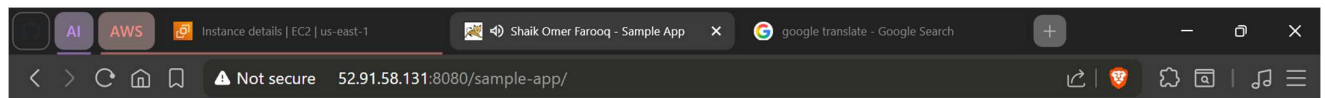
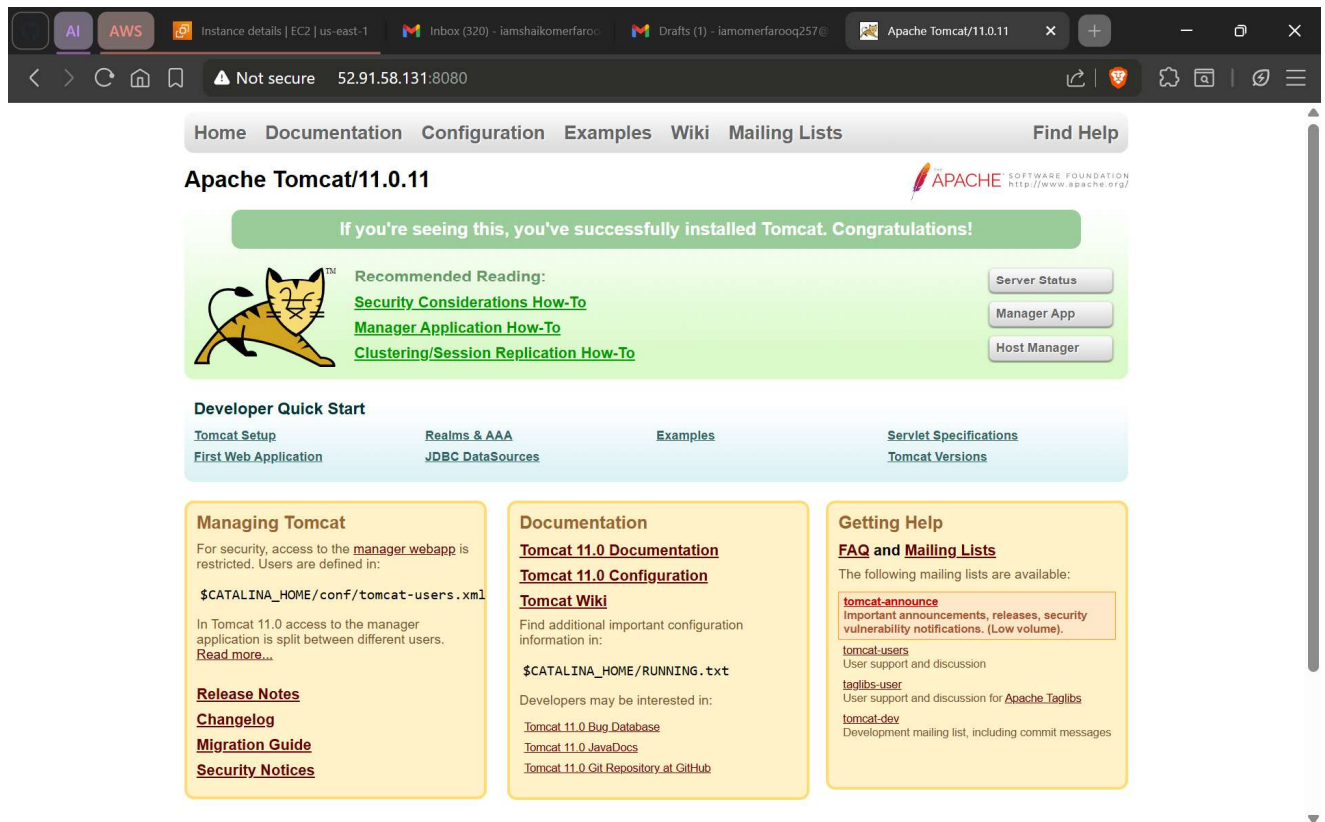
9. Follow the on-screen instructions until the installation is complete.

Installing Tomcat:

1. Using Windows Service Installer Download Tomcat: Visit the [official Apache Tomcat website](https://tomcat.apache.org/whatsnew.html) and download the Windows Service Installer for Tomcat 11.
2. Run the Installer by launching the .exe file. Follow the setup wizard, accept the license, and configure ports – HTTP (default: 8080) and admin credentials.
3. Set Java Path: **Provide the path to your JDK installation** (auto-detected if JAVA_HOME is set).
4. Complete Installation: Click "**Install**" and then "**Finish**." Optionally, check "**Run Apache Tomcat**" to start the service immediately.

Check Tomcat Service and Deploy Sample App:

1. To check if the Tomcat service is running, copy the public IPv4 address of the instance and open it with the designated port number 8080.
<http://98.88.84.59:8080>
2. The output is attached below
 1. The deployment location of Tomcat in the Windows Server is **C:\Program Files\Apache Tomcat\Tomcat11\webapps**
 2. Either you can download a sample app and deploy it directly. But what's fun in that? So, I made a sample custom app and deployed it.
 3. The output of the sample apps is attached below.



Hallo, ich bin Shaik Omer Farooq.

Alle, alle. Ich bin für alle da. Dominiere alle. Ich bin der Champion.

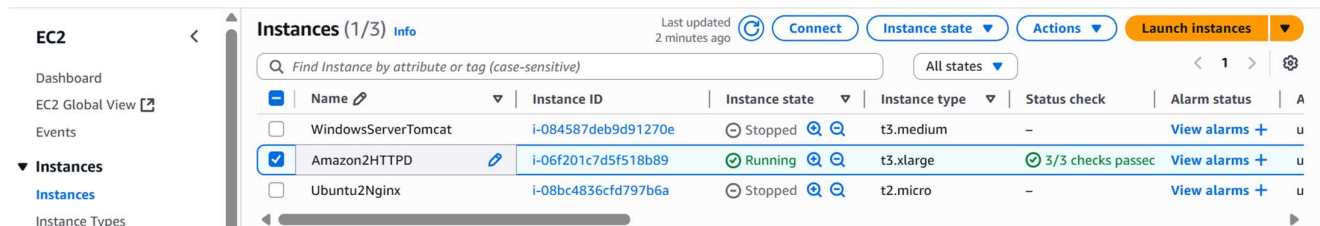
Music Player



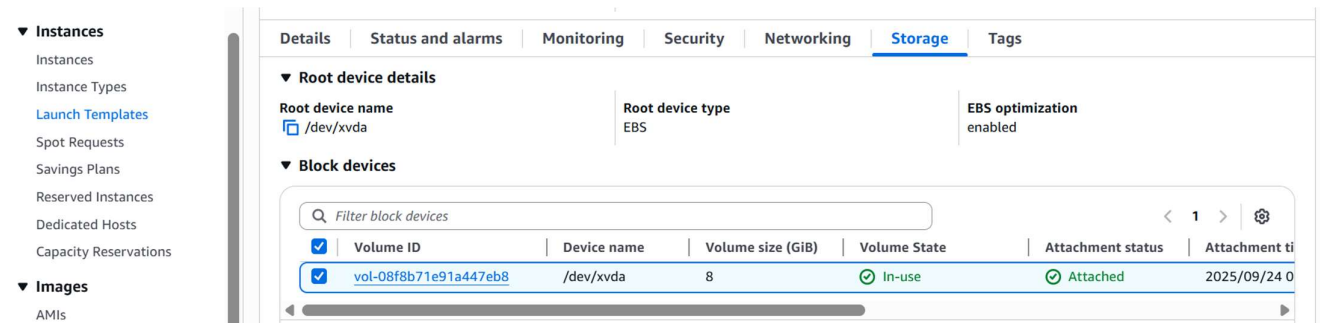
4. Take a snapshot of the instance created in Task 1.

Steps to take a snapshot of the instance that has HTTPD using EBS (Elastic Block Store).

1. Log in to the AWS console and navigate to **EC2**. Under EC2, look for “**Instances**”, and look for the instance that has HTTPD.

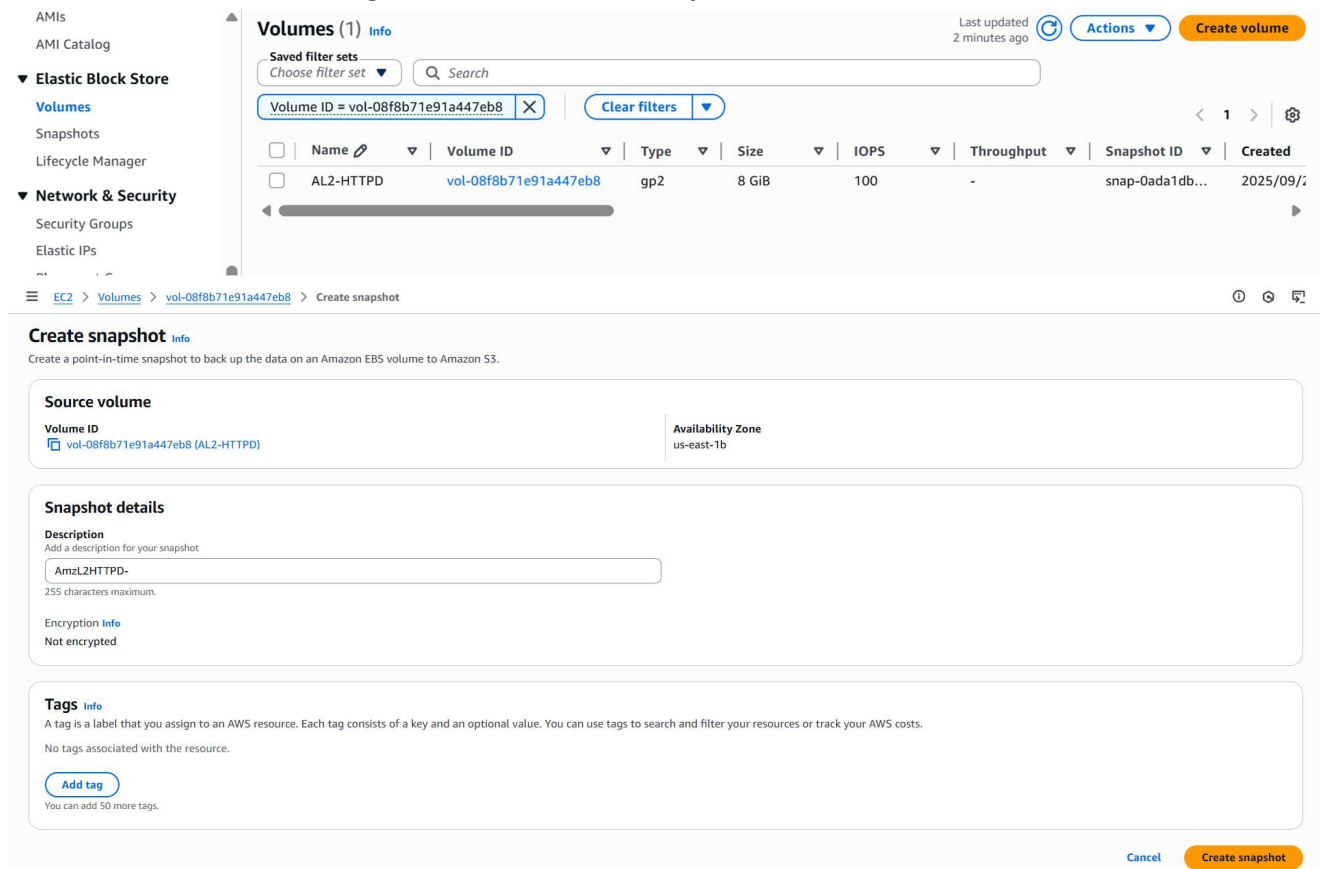


2. Click on the instance ID of the HTTPD instance, and scroll down and click on “Storage” options.



3. Identify and select the EBS volume you want to back up.

4. With the volume selected, go to **Actions > Create Snapshot**.



5. Provide a name and description for your snapshot. You can also choose to add tags. Click on “**Create Snapshot**”. Check the newly created snapshot in the “Snapshots” tab.

5. Assign a password-less authentication for the EC2 created in Task 2.

Steps to assign password-less authentication for the EC2 Instance running on Ubuntu AML:

1. Generate a local key with:

\$ ssh-keygen

2. You can name the key if you wish to, or leave it as it is. A default name will be given to the file.

3. When we generate a key, two files will be created. One will be a public key, and another will be a private key.

4. Open the public key from the generated location. Copy the key and keep it ready to paste into the remote system.

```
iamom@Omer MINGW64 ~/.ssh
$ ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/iamom/.ssh/id_ed25519):
Enter passphrase for "/c/Users/iamom/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/iamom/.ssh/id_ed25519
Your public key has been saved in /c/Users/iamom/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:Xdk/puC6BsnONXpmonIAeLVXYc8+hs4+S+PCmwVr4ik iamom@Omer
The key's randomart image is:
+--[ED25519 256]--+
|      o.          |
|    . . . +      |
|  . . . . =      |
| . o . o . = .   |
| o . oSoo + .    |
|  . * =o.. . .   |
|  O X o*  O      |
|  E B X=.o o     |
|  . =o*=o.       |
+-----[SHA256]-----+

iamom@Omer MINGW64 ~/.ssh
$ ls
id_ed25519  id_ed25519.pub

iamom@Omer MINGW64 ~/.ssh
$ cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIPZA13VA3aUnzAcJOth2mZCNZEhb+Tr5eQe/1m2OHBVI iamom@Omer
```

5. Connect to the remote system using SSH.

\$ ssh -i <pem_key> <user_name>@<Public_IP>

6. Open the "authorized_keys" file in "/home/ec2-user/.ssh" directory.

\$ cd /home/ec2-user/.ssh (or) \$ cd /home/ubuntu/.ssh

\$ vi authorized_keys -> Paste the local public key here

```
iamom@Omer MINGW64 ~/Downloads
$ ssh -i ec2.pem ec2-user@34.228.22.217
The authenticity of host '34.228.22.217 (34.228.22.217)' can't be established.
ED25519 key fingerprint is SHA256:EGVMYfyY6qfZ4BWhFgNOD/JXHRNstM6yTHGkzjIePBC.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '34.228.22.217' (ED25519) to the list of known hosts.
Last login: Wed Sep 24 10:07:06 2025 from 103.143.169.218

#_
~\##### Amazon Linux 2
~\##### AL2 End of Life is 2026-06-30.
~\#####
~\#####
~\##### A newer version of Amazon Linux is available!
~\##### Amazon Linux 2023, GA and supported until 2028-03-15.
~\##### https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-172-31-16-85 ~]$ cd .ssh
[ec2-user@ip-172-31-16-85 .ssh]$ ls
authorized_keys
[ec2-user@ip-172-31-16-85 .ssh]$ vi authorized_keys
```



```
$ chmod 700 /home/ec2-user/.ssh && $ chmod 600 /home/ec2-user/.ssh/authorized_keys
$ chown ec2-user:ec2-user /home/ec2-user/.ssh -R
```

8. Switch to root and change the make changes in the **"sshd_config"** file which is located in the directory **"/etc/ssh"**. Type **'yes'** after the **PasswordAuthentication**.

```
# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no
```

10. If all the steps are correct, the output will look something like the image below.

```

iamom@Omer MINGW64 ~/Downloads
$ ssh ec2-user@34.228.22.217
Last login: wed Sep 24 10:32:20 2025 from 103.143.169.218

#_
~\_ #####_      Amazon Linux 2
~~~\_#####\
~~~\_####|      AL2 End of Life is 2026-06-30.
~~~\_#/
~~~\_#/'-'>
~~~~
~~~~_._/
~~~~_/_/
~~~~_/_/m/'-'

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-172-31-16-85 ~]$ |

```

Steps to Launch an EC2 Instance Using the Spot Purchase Option.

- aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 7348-4675-3465

omerfarooq257

EC2 > Instances > Launch an instance

Purchasing option | Info

☐ None

☐ Capacity Blocks
Launch instances for your active capacity blocks

☒ Spot instances
Request Spot Instances at the Spot price, capped at the On-Demand price

Discard Spot instance options

Spot Instance Options | Info

Specify Spot Instance Options such as Maximum Price, Request type, expiration date and interruption behavior

Maximum price | Info

☐ No maximum price
Request Spot Instances at the Spot price, capped at the On-Demand price

☒ Set your maximum price (per instance/hour)

\$ 1.5

Request type | Info

One-time

Valid to | Info

☒ No request expiry date
The default value is no expiry date

☐ Set your request expiry date

Interruption behavior | Info

Select

Capacity reservation | Info

▼ Summary

Number of instances | Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.8.2...read more
ami-08982f1c5bf9d9d76

Virtual server type (Instance type)
t2.micro

Firewall (security group)
MySecurityGroup

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier

Cancel

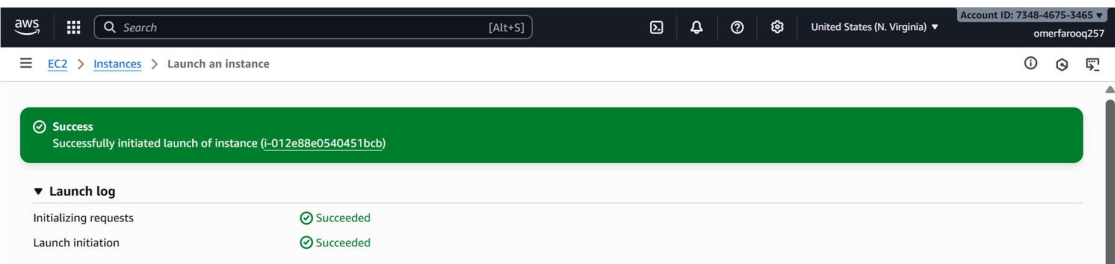
Launch instance

Preview code

CloudShell

Feedback

© 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences



```
i@mom@Omer MINGW64 ~/Downloads
$ ssh -i ec2.pem ec2-user@52.91.91.88
The authenticity of host '52.91.91.88 (52.91.91.88)' can't be established.
ED25519 key fingerprint is SHA256:owpfk3a0KRV+GEj7qu7ykpvcfJ2vpC9ESC/c2WKm52Q.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '52.91.91.88' (ED25519) to the list of known hosts.
```

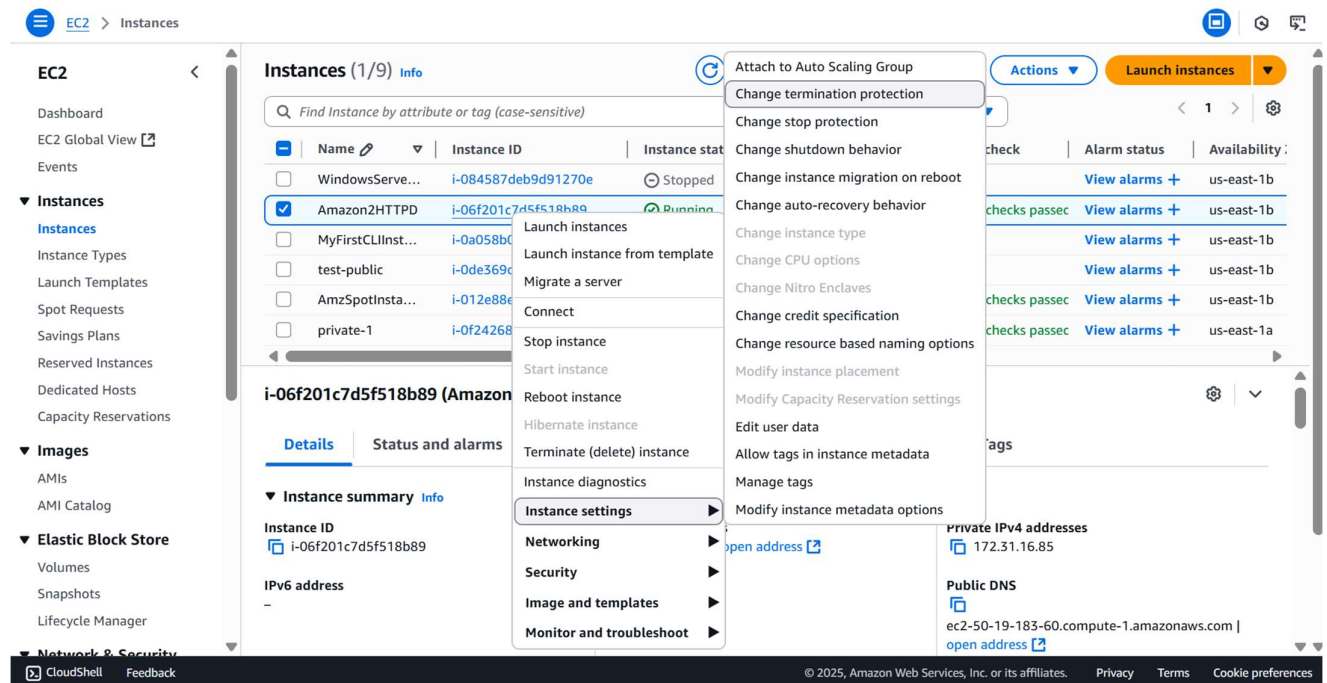
```
#_
~\#####      Amazon Linux 23
~~\#####\
~~\####|
~~\#/
~~V~'-> https://aws.amazon.com/linux/amazon-linux-2023
~~~~
~~.-.-.-
~/m/'
```

```
[ec2-user@ip-172-31-23-248 ~]$ |
```

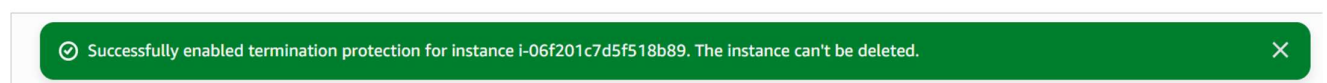
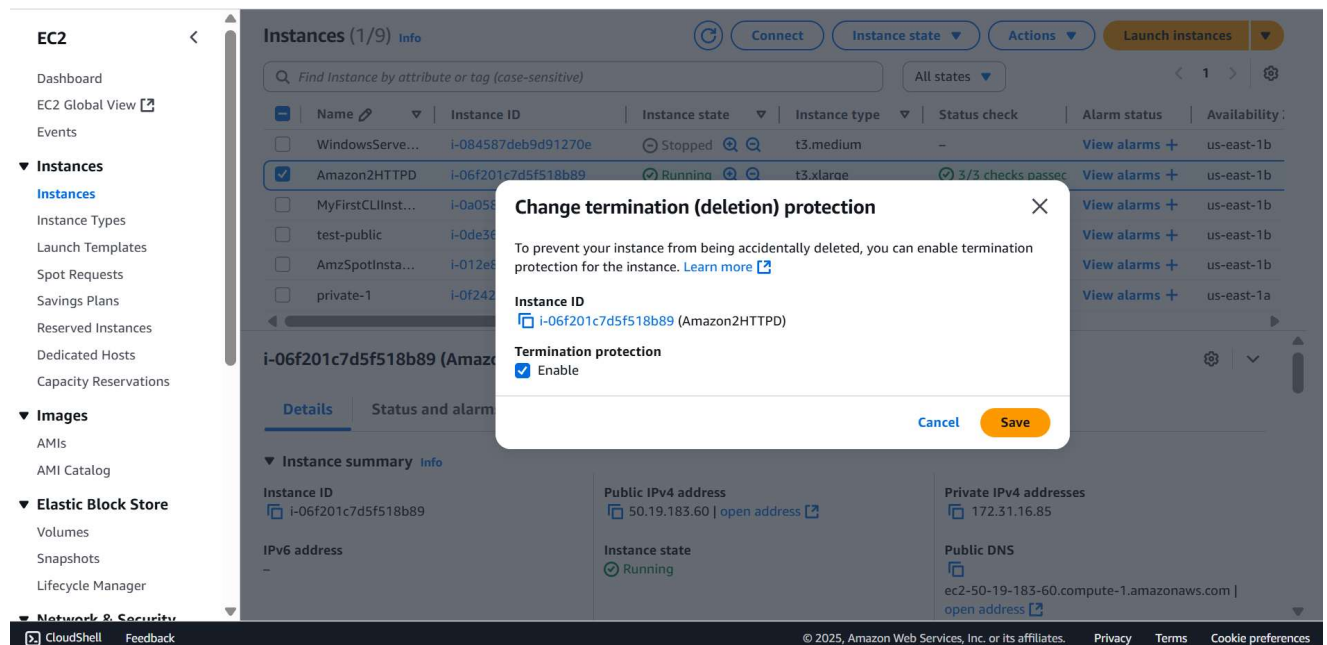

7. Enable Termination policy on the EC2 created in Task 2.

Steps to enable termination policy on an already launched instance.

1. Go to **EC2 Dashboard** → **Instances**. Select your Ubuntu instance from Task 2.
2. Right-click on the instance (or use the Actions menu). Select **"Instance settings"** → **"Change termination protection"**.



3. Check the "Enable" checkbox. Click "Save".

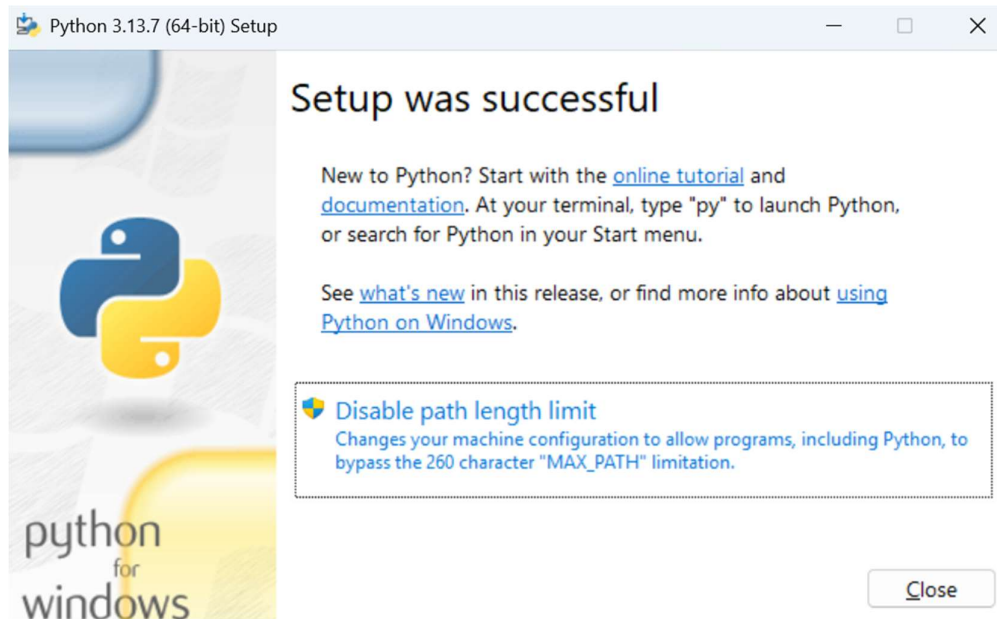


8. Launch one EC2 using AWS CLI.

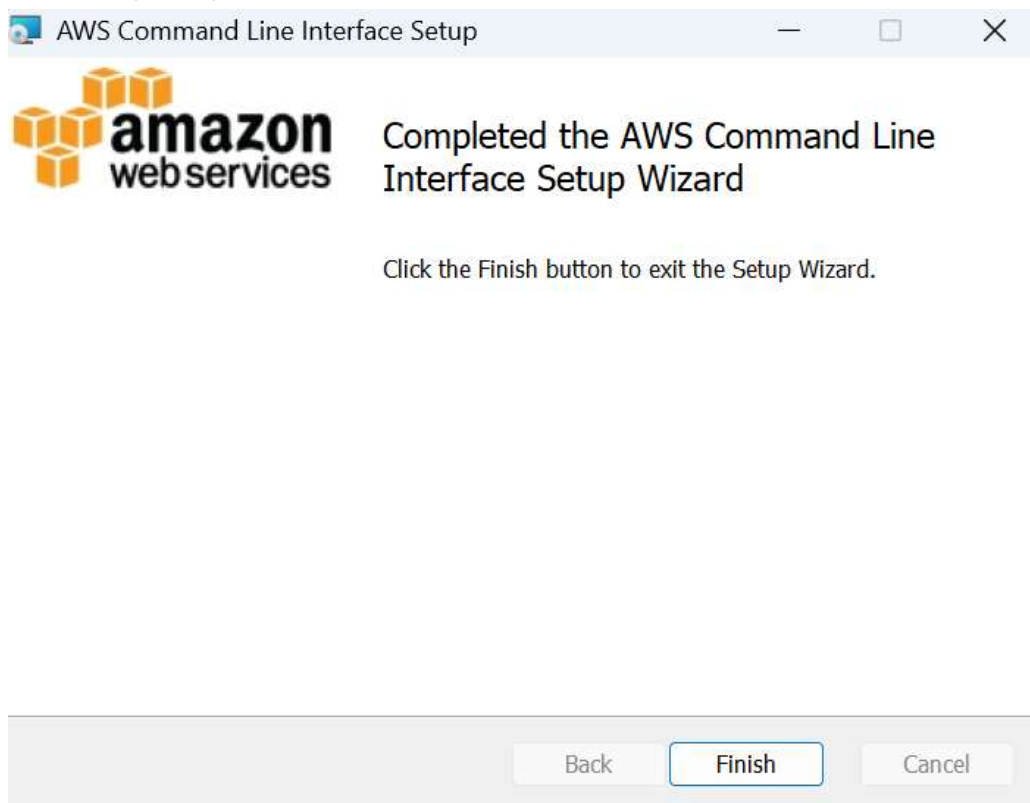
Steps to launch an EC2 Instance using AWS CLI.

Follow the procedure mentioned in the link: <https://docs.aws.amazon.com/cli/v1/userguide/cli-services-ec2-instances.html>

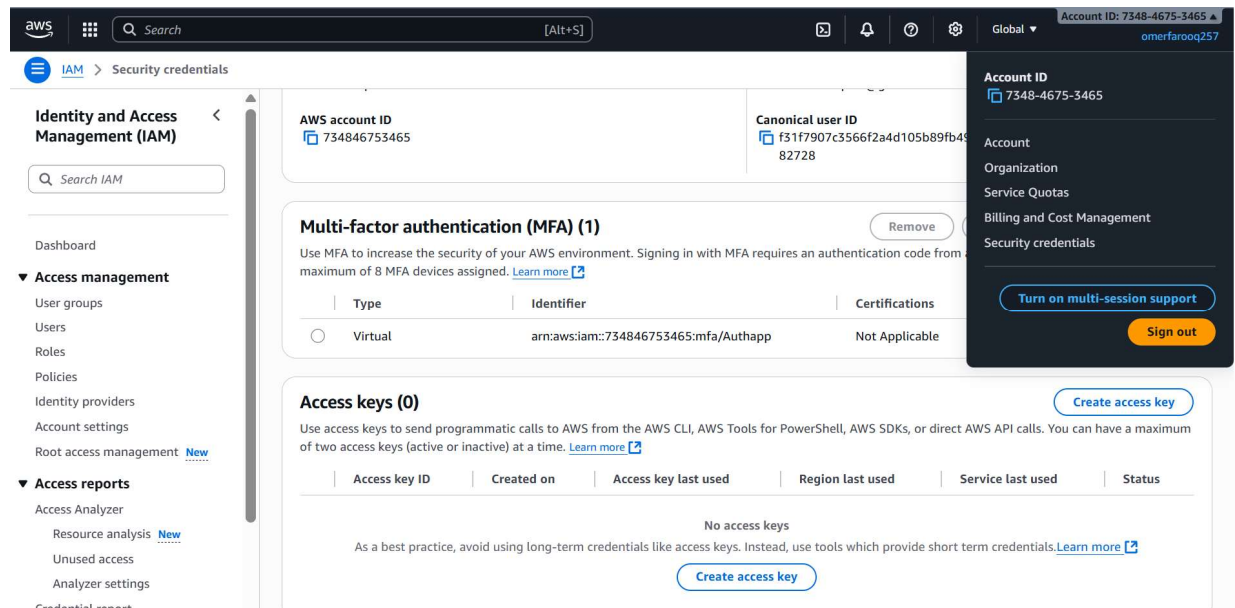
1. The AWS CLI version 1 is built using the SDK for Python, and therefore requires you to install a compatible version of Python. Search for 'Python for Windows' and download and install the latest version of Python.



2. Download and install the appropriate AWS CLI MSI installer. I installed the AWS CLI MSI installer for Windows (64-bit).



3. Log in to the AWS Console. Navigate to Security Credentials; you'll find it by clicking on your username in the right corner of the console.
4. Scroll down and find the Access Keys Section. You'll see a table that might be empty if this is your first time.



Alternatives to root user access keys [Info](#)

⚠ Root user access keys are not recommended

We don't recommend that you create root user access keys. Because you can't specify the root user in a permissions policy, you can't limit its permissions, which is a best practice.

Instead, use alternatives such as an IAM role or a user in IAM Identity Center, which provide temporary rather than long-term credentials. [Learn More](#)

If your use case requires an access key, create an IAM user with an access key and apply least privilege permissions for that user. [Learn More](#)

Continue to create access key?

- ☒ I understand creating a root access key is not a best practice, but I still want to create one.

[Cancel](#)

[Create access key](#)

5. Click the "Create New Access Key" button. You'll see a warning pop-up about root access keys - click "Continue" for learning purposes. A dialog box will appear with your new credentials.
6. This is the **ONLY time you'll ever see the Secret Access Key!** AWS will never show it to you again. Save the keys or download them.
7. Configure AWS CLI in Git Bash. Use the following command:
\$ aws configure -> Enter the details (Configure AWS CLI in Git Bash, AWS Secret Access Key, Default region name, and Default output format).

```

iamom@Omer MINGW64 ~/Downloads
$ aws --version
aws-cli/1.42.37 Python/3.12.6 windows/11 botocore/1.40.37

iamom@Omer MINGW64 ~/Downloads
$ aws configure
AWS Access Key ID [None]: AKIA2WGCHTK42GJLMNKJ
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]: json

iamom@Omer MINGW64 ~/Downloads
$ aws configure list

```

Name	Value	Type	Location
profile	<not set>	None	None
access_key	*****MNKJ	shared-credentials-file	
secret_key	*****Nyps	shared-credentials-file	
region	us-east-1	config-file	~/.aws/config

8. Check if the configuration is done right with the command:

\$ aws configure list

9. Start the launching process using the command/script:

```

iamom@Omer MINGW64 ~/Downloads
$ aws ec2 describe-images \
  --owners amazon \
  --filters \
    "Name=name,Values=amzn2-ami-hvm-*" \
    "Name=state,Values=available" \
    "Name=architecture,Values=x86_64" \
  --query 'Images | sort_by(@, &CreationDate) | [-1].{ImageId:ImageId,Name:Name,CreationDate:CreationDate}' \
  --output table

```

DescribeImages	
CreationDate	2025-09-10T19:04:06.000Z
ImageId	ami-0254b2d5c4c472488
Name	amzn2-ami-hvm-2.0.20250915.0-x86_64-gp2

i. Find the Right AMI (Operating System Image)

Find the latest Amazon Linux 2 AMI

**aws ec2 describe-images **

**--owners amazon **

**--filters **

**"Name=name,Values=amzn2-ami-hvm-*" **

**"Name=state,Values=available" **

**"Name=architecture,Values=x86_64" **

**--query 'Images | sort_by(@, &CreationDate) | [-1].{ImageId:ImageId,Name:Name,CreationDate:CreationDate}' **

--output table

Copy the ImageId from the output - you'll need it!

Example: ami-0abcdef1234567890


```
ec2-user@ip-172-31-22-166:~  
iamom@Omer MINGW64 ~/Downloads  
$ # Create security group and capture the Group ID  
SECURITY_GROUP_ID=$(aws ec2 create-security-group \  
--group-name my-first-sg \  
--description "My first security group for learning" \  
--query 'GroupId' \  
--output text)  
  
echo "Created Security Group: $SECURITY_GROUP_ID"  
  
# Add SSH access rule (port 22)  
aws ec2 authorize-security-group-ingress \  
--group-id $SECURITY_GROUP_ID \  
--protocol tcp \  
--port 22 \  
--cidr 0.0.0.0/0  
  
# Add HTTP access rule (port 80) - optional, for web servers  
aws ec2 authorize-security-group-ingress \  
--group-id $SECURITY_GROUP_ID \  
--protocol tcp \  
--port 80 \  
--cidr 0.0.0.0/0  
  
echo "Security group rules added successfully!"  
Created Security Group: sg-0c842ac5c157e83f8  
{  
  "Return": true,  
  "SecurityGroupRules": [  
    {  
      "SecurityGroupRuleId": "sgr-0945ab9b2c9287a16",  
      "GroupId": "sg-0c842ac5c157e83f8",  
      "GroupOwnerId": "734846753465",  
      "IsEgress": false,  
      "IpProtocol": "tcp",  
      "FromPort": 22,  
      "ToPort": 22,  
      "CidrIpv4": "0.0.0.0/0",  
      "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:734846753465:security-group-rule/sg-0c842ac5c157e83f8:sgr-0945ab9b2c9287a16"  
    }  
  ]  
}  
  
"Return": true,  
"SecurityGroupRules": [  
  {  
    "SecurityGroupRuleId": "sgr-09b7e56baa35c95a0",  
    "GroupId": "sg-0c842ac5c157e83f8",  
    "GroupOwnerId": "734846753465",  
    "IsEgress": false,  
    "IpProtocol": "tcp",  
    "FromPort": 80,  
    "ToPort": 80,  
    "CidrIpv4": "0.0.0.0/0",  
    "SecurityGroupRuleArn": "arn:aws:ec2:us-east-1:734846753465:security-group-rule/sg-0c842ac5c157e83f8:sgr-09b7e56baa35c95a0"  
  }  
]  
}  
Security group rules added successfully!  
iamom@Omer MINGW64 ~/Downloads
```

ii. Create a Security Group (Firewall)

Create security group and capture the Group ID

```
SECURITY_GROUP_ID=$(aws ec2 create-security-group \  
--group-name my-first-sg \  
--description "My first security group for learning" \  
--query 'GroupId' \  
--output text)
```

--output text)

echo "Created Security Group: \$SECURITY_GROUP_ID"

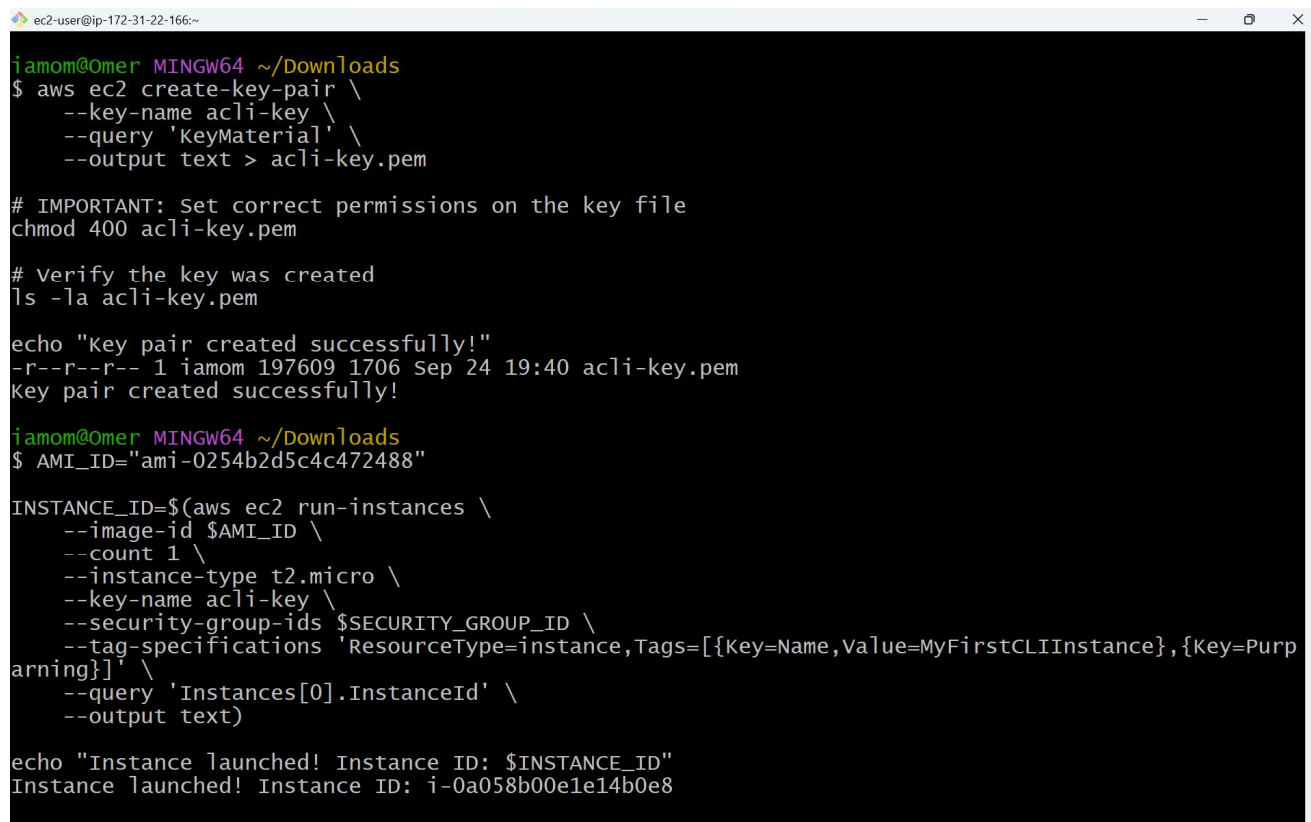
Add SSH access rule (port 22)

```
aws ec2 authorize-security-group-ingress \  
  --group-id $SECURITY_GROUP_ID \  
  --protocol tcp \  
  --port 22 \  
  --cidr 0.0.0.0/0
```

Add HTTP access rule (port 80) - optional, for web servers

```
aws ec2 authorize-security-group-ingress \  
  --group-id $SECURITY_GROUP_ID \  
  --protocol tcp \  
  --port 80 \  
  --cidr 0.0.0.0/0
```

echo "Security group rules added successfully!"

A terminal window titled 'ec2-user@ip-172-31-22-166:~' with a dark background. The user 'iamom@Omer' is in a 'MINGW64' shell at '~/Downloads'. The terminal shows the following commands and output:
1. `aws ec2 create-key-pair \`
 `--key-name acli-key \`
 `--query 'KeyMaterial' \`
 `--output text > acli-key.pem`
2. `# IMPORTANT: Set correct permissions on the key file`
 `chmod 400 acli-key.pem`
3. `# Verify the key was created`
 `ls -la acli-key.pem`
4. `echo "Key pair created successfully!"`
 Output: `-r--r--r-- 1 iamom 197609 1706 Sep 24 19:40 acli-key.pem`
 `Key pair created successfully!`
5. `iamom@Omer MINGW64 ~/Downloads`
 `$ AMI_ID="ami-0254b2d5c4c472488"`
6. `INSTANCE_ID=$(aws ec2 run-instances \`
 `--image-id $AMI_ID \`
 `--count 1 \`
 `--instance-type t2.micro \`
 `--key-name acli-key \`
 `--security-group-ids $SECURITY_GROUP_ID \`
 `--tag-specifications 'ResourceType=instance,Tags=[{Key=Name,Value=MyFirstCLIInstance},{Key=Purp`
 `arning}]' \`
 `--query 'Instances[0].InstanceId' \`
 `--output text)`
7. `echo "Instance launched! Instance ID: $INSTANCE_ID"`
 Output: `Instance launched! Instance ID: i-0a058b00e1e14b0e8`

iii. Create a Key Pair for SSH Access

Create key pair and save private key

```
aws ec2 create-key-pair \  
  --key-name my-learning-key \  
  --query 'KeyMaterial' \  
  --output text > my-learning-key.pem
```


IMPORTANT: Set correct permissions on the key file

chmod 400 my-learning-key.pem

Verify the key was created

ls -la my-learning-key.pem

echo "Key pair created successfully!"

iv. Launch Your EC2 Instance

Set your AMI ID (replace with the ID you found earlier)

AMI_ID="ami-0abcdef1234567890" # Replace this with your actual AMI ID

Launch the instance and capture the Instance ID

**INSTANCE_ID=\$(aws ec2 run-instances **

**--image-id \$AMI_ID **

**--count 1 **

**--instance-type t2.micro **

**--key-name my-learning-key **

**--security-group-ids \$SECURITY_GROUP_ID **

--tag-specifications

'ResourceType=instance,Tags=[{Key=Name,Value=MyFirstCLIInstance},{Key=Purpose,Value=Learning}]

**' **

**--query 'Instances[0].InstanceId' **

--output text)

echo "Instance launched! Instance ID: \$INSTANCE_ID"

```
iamom@Omer MINGW64 ~/Downloads
$ aws ec2 describe-instances \
  --instance-ids $INSTANCE_ID \
  --query 'Reservations[0].Instances[0].{InstanceId:InstanceId,State:State.Name,PublicIP:PublicIpAddress,PrivateIP:PrivateIpAddress,InstanceType:InstanceType}' \
  --output table
```

DescribeInstances	
InstanceId	i-0a058b00e1e14b0e8
InstanceType	t2.micro
PrivateIP	172.31.22.166
PublicIP	52.91.247.126
State	running

v. Get the Public IP Address

Get instance details including public IP

**aws ec2 describe-instances **

**--instance-ids \$INSTANCE_ID **

--query

**'Reservations[0].Instances[0].{InstanceId:InstanceId,State:State.Name,PublicIP:PublicIpAddress,PrivateIP:PrivateIpAddress,InstanceType:InstanceType}' **

--output table

```
iAmom@Omer-MINGW64 ~/Downloads  
$ ssh -i ac1i-key.pem ec2-user@52.91.247.126  
The authenticity of host '52.91.247.126 (52.91.247.126)' can't be established.  
ED25519 key fingerprint is SHA256:/rnVyG71D3K59FhgiYXG6gWQNYP3KdQOGcdki/3w8aI.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '52.91.247.126' (ED25519) to the list of known hosts.  
  
_#  
~\##### Amazon Linux 2  
~~\#####\  
~~\###| AL2 End of Life is 2026-06-30.  
~~\#/ \  
~~V~'->  
~~~~  
~~~~  
~~~~  
~/m/'-/-/  
  
A newer version of Amazon Linux is available!  
  
Amazon Linux 2023, GA and supported until 2028-03-15.  
https://aws.amazon.com/linux/amazon-linux-2023/  
  
[ec2-user@ip-172-31-22-166 ~]$ ls  
[ec2-user@ip-172-31-22-166 ~]$ pwd  
/home/ec2-user
```

10. Once the configuration and launching of the instance is done, and an IP is acquired, connect to your Instance via SSH.

```
$ ssh -i acli-key.pem ec2-user@52.91.247.126
```

11. Explore the new Instance and check if it's working fine.

```
$ uname -a && df -h && free -m
```